

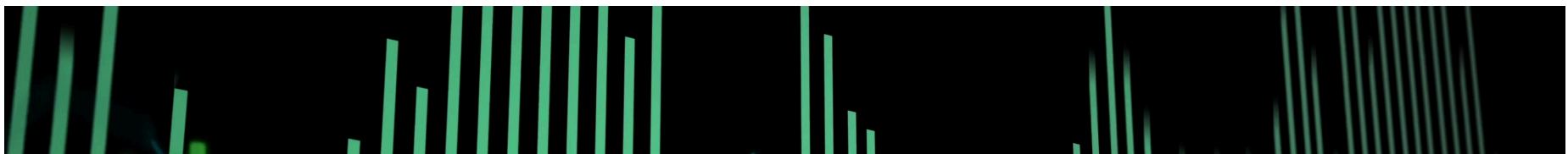
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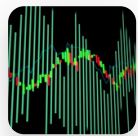
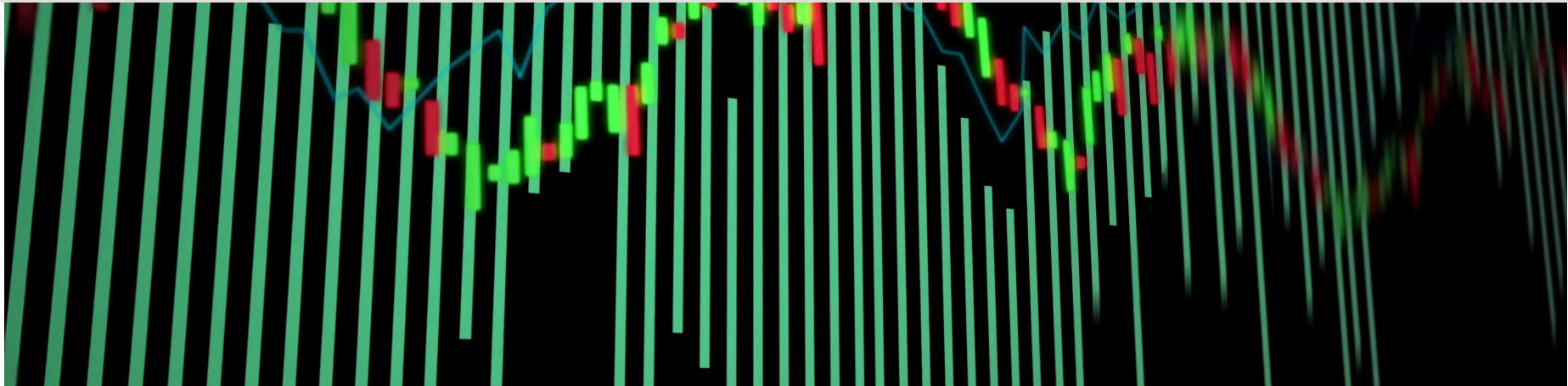
The Quadrillion-Dollar Disagreement on AI and the Economy

AI forecasts span a range of potential futures, from economic stagnation to explosive growth. The divergence traces to three specific assumptions—each generating predictions we can already test.

May 11, 2026

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“The Quadrillion-Dollar Disagreement on AI and the Economy” by Anton Shenk

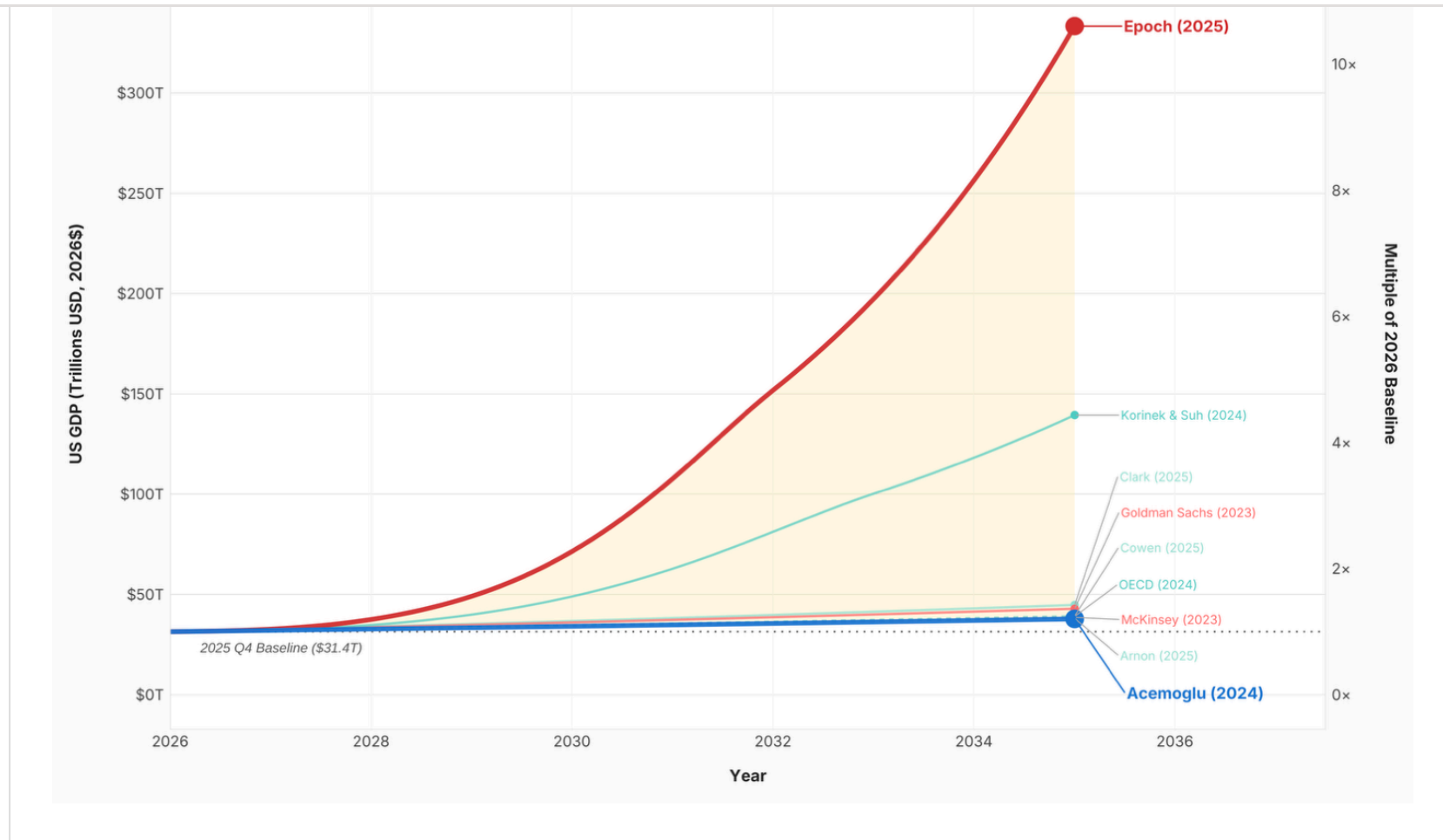
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In the three years since OpenAI launched ChatGPT, economists and AI researchers have published forecasts projecting that, over the next decade, AI will add to annual growth by amounts ranging from as little as 0.1% to as much as 30%. By 2035, the gap between these forecasts nears a quadrillion dollars: an amount that exceeds a decade’s worth of current global output.

The Quadrillion-Dollar Delta

Projected US GDP Under Alternative AI Growth Estimates, 2026-2035



Source: [2025 Q4 US GDP](#) from Federal Reserve Economic Data. Estimates sourced from [Goldman Sachs \(2023\)](#), [McKinsey \(2023\)](#), [Acemoglu \(2024\)](#), [OECD \(2024\)](#), [Korinek & Suh \(2024\)](#), [Arnon \(2025\)](#), [Cowen \(2025\)](#), [Clark \(2025\)](#), and [Epoch AI \(2025\)](#). Not all estimates model US GDP directly: Epoch AI models Gross World Product; McKinsey models productivity across 47 countries; Korinek & Suh calibrate to advanced-economy baselines. Growth rates from these sources are applied to the US GDP baseline for comparability. Where ranges are reported, the median is used. Where excess growth is reported rather than levels, a baseline 2% GDP growth is assumed.

describe civilizational transformation through explosive growth, wholesale economic restructuring, and unprecedented governance challenges. No treasury, central bank, nor legislature can prepare effectively for such a wide range of outcomes.

Yet the machinery of government does not pause for epistemic crises. Budgets are being written, pension obligations set, tax codes revised. Knowingly or not, each policymaking decision places an implicit wager on which future is most likely to materialize.

Betting on the wrong forecast for AI's economic impact would carry severe consequences. If

governments prepare for explosive growth that never arrives—borrowing against imagined AI windfalls to fund infrastructure and expand social programs—they risk sovereign debt crises and spiraling inflation when tax revenues disappoint. If they prepare for modest growth and explosive transformation arrives instead—maintaining current unemployment systems, underfunding retraining programs, and leaving regulatory frameworks unchanged—societies may face mass labor displacement, entrenched inequality, and unprecedented policy challenges, all of which can foster existential political instability.

This divergence in forecasts stems from three testable disagreements about how AI interacts with the economy: (1) whether AI's jagged capability profile (excelling at some complex tasks while failing at seemingly simpler ones) represents temporary limitations or fundamental constraints on automation; (2) whether institutional friction limits technological potential, and (3) whether AI can automate innovation itself.

Instead of guessing about an AI-influenced future, we should analyze which assumptions are holding in the real economy. This article maps the three core questions driving the quadrillion-dollar delta—and shows what data policymakers should watch to better predict which world we'll enter.

The first disagreement sounds simple: what kinds of work can AI actually do?

MIT economist Daron Acemoglu projects that AI will boost US productivity by 0.71% over 10 years. University of Virginia economist Anton Korinek’s scenarios reach 18% annual GDP growth. Both Acemoglu and Korinek are leading scholars who published their estimates through the National Bureau of Economic Research, the most prestigious working-paper series in economics. Acemoglu won the Nobel Prize in 2024; in 2025, Korinek made TIME’s 2025 list of the 100 most influential people in AI. The 25-fold difference in their projections is not a disagreement about whether AI is impressive—each acknowledges that the technology is already impactful. It is a disagreement about whether AI’s current limitations are permanent, or will scale into the complex, context-dependent work where most economic value resides.

The Skeptical Case: AI Faces a “Hard Tasks” Wall

Acemoglu starts by focusing on the qualitative differences between tasks in the economy. An AI system that writes fluent marketing copy has automated a real job—but copywriters represent roughly 0.17% of the US workforce. However, the remaining 99.83% of jobs may still require skills that AIs are unable to do.

Health care workers—nurses, home health aides, medical assistants—represent roughly 18% of US GDP and 13% of total US employment. Combined with those serving in education and the skilled trades, these employees—whom Acemoglu calls “hard task” workers—constitute a commanding share of jobs and national output. In Acemoglu’s view, these jobs resist automation: they are context-dependent, socially embedded, and lack clear metrics for success.

audio, standardizing a dataset—with such tasks, you know what “done well” looks like, and the steps to get there are straightforward. AI learns these quickly.

Hard tasks resist easy verification. Diagnosing a patient’s cough, hiring the right candidate, teaching a classroom of teenagers: the desired outcome depends on vast contextual factors, and whether the task was performed well may not be immediately evident. Without clear success criteria, AI can only mimic average human behavior rather than exceed it. Acemoglu argues that, even under optimistic assumptions about AI capabilities, AI exposure is limited to just 5% of output.

AI automation will displace labor, not replace it. As AI automates “easy” tasks, Acemoglu hypothesizes, labor won’t disappear; it will refocus on “hard” tasks toward which AI contributes little. In this model, even as call centers become more efficient and data processing accelerates, economy-wide productivity barely moves, because labor drifts into healthcare, education, and the skilled trades. In aggregate, the impact of AI on labor resembles squeezing a balloon: visible displacement but no overall contraction.

This is not hypothetical. It is exactly what happened during the first wave of personal computers. Spectacular technological capability produced only modest aggregate productivity gains, because the technology was applied to a narrow slice of actual work.

The Explosive Case: Difficult Tasks Can Be Surmounted

Korinek rejects this binary, arguing that task complexity is a distribution rather than a wall. Some tasks are easier, some are harder, but difficulty is a continuous spectrum: verifying an audio transcription is easier than verifying a medical diagnosis but much harder than verifying numerical calculations. While Acemoglu assumes

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This distinction is crucial. If task difficulty is a continuum and AI capabilities continue to scale with marginal compute and data—following the long-standing “scaling laws” observed in AI development—then “hard” tasks just require more scale. The question is not whether AI automates legal research or medical diagnosis, it’s when it will do so. The answer, history suggests, is sooner than most expect.

Labor-market data may illuminate AI’s emerging impact. The empirical battleground is already visible. Can AI handle the tasks Acemoglu classifies as structurally hard? Evidence is mixed and revealing. AI passes the US Medical Licensing and bar exams, each of which are supposedly hard tasks. Yet current AI and robotics systems struggle with physical tasks such as folding laundry and washing dishes—tasks any teenager can learn. More tellingly, adoption surveys show businesses hesitating to scale AI across their operations, revealing preferences about reliability that marketing materials obscure.

While AI benchmarks serve as proxies for the breadth of AI automation, this question will ultimately be settled in labor markets. If copywriter wages fall while nurses, teachers, and tradespeople remain valuable, Acemoglu is right: AI is automating “easy” tasks but not “hard” ones. But if wage gaps contract across occupations as AI adoption expands—faster than retraining and relicensing alone could explain—then the scaling hypothesis is eating through task complexity faster than skeptics believed possible.

Companies Produce?

AI capabilities are necessary but not sufficient for broad impact and automation. Even if AI can perform a task, deploying it at scale lies behind what technologists call “schlep”—the unglamorous work of implementation—such as navigating regulation, liability, and IT integration.

Goldman Sachs projects that AI will add 1.5% to annual productivity growth over a decade. The Organisation for Economic Co-operation and Development (OECD) models barely break 0.5%. Each organization acknowledges that AI performs useful work, yet they disagree about whether the economy can actually absorb that work at scale.

The Skeptical Case: Institutions Are Stickier Than Markets

OECD economists base their bearish forecast on economic friction. AI adoption is impeded by a thicket of regulations, licensing, liability, and inertia—all of which move at legislative, not technological, speed.

Consider health care, which represents roughly 10% of GDP in advanced economies. AI reads medical images more accurately than most radiologists. Yet deploying it requires navigating medical device regulations, malpractice liability, insurance reimbursement codes, and state credentialing boards. The EU’s AI Act classifies medical AI as “high-risk,” triggering extensive compliance requirements. US liability law falls short of adjudicating a new class of uncertainty: who pays when the algorithm misses a tumor? Medical boards move at the speed of committee consensus.

The result of this regulatory friction is spectacular lab performance, but glacial adoption.

because they resist automation. If AI accelerates manufacturing and IT but barely touches health care and education, aggregate productivity stays muted, because the economy keeps reallocating toward the resistant sectors. The economy optimizes the shrinking parts, while the growing parts stay stuck.

The Explosive Case: Markets Fund What Works

Goldman Sachs rejects the institutional friction narrative, treating AI as a General Purpose Technology (GPT) comparable to electricity or the combustion engine. The mechanism is capital deepening: firms see ROI, capital markets fund adoption, workers upskill, complementary technologies emerge, and productivity compounds.

The timeline for integrating other transformative technologies spanned decades: electrification took 40 years, while computerization took 20. Unlike past GPTs, AI may also accelerate its own adoption, automating compliance and streamlining regulatory approvals in ways that electricity and steam never could. If technology generates returns, capital clears obstacles. Regulatory frameworks adapt. Firms reorganize. What earns profits gets deployed.

The friction these models acknowledge is economic, not structural. The binding constraint on AI adoption is investment capacity and worker adjustment speed—not institutions.

While each predicts AI proliferation, these scenarios differ on how and when. The difference between the two cases is whether the binding constraint is technological or institutional. If friction dominates, the challenge is not R&D funding but regulatory reform and workforce retraining. The question shifts from “Can AI do the task?” to “Will we let it?”

integration is accelerating beyond tech and financial services into regulated, labor-intensive sectors like health care. If it is, institutional barriers are proving permeable. If adoption plateaus in sectors where regulation, licensing, and liability bind, friction is dominating.

Question 3: Can AI Automate Innovation Itself?

Beyond disagreements over direct automation, economists also disagree over whether AIs can automate discovery: the key input that makes everything else in the economy possible.

The Skeptical Case: AI Will Provide a One-Time Economic Boost

Economists Martin Baily, Erik Brynjolfsson, and Anton Korinek’s [2023 survey](#) provides the framework for understanding this position, even as the authors themselves argue for more transformative outcomes: AI is a **General Purpose Technology (GPT)**—broadly applicable, generating cross-sector spillovers, raising productivity of capital and labor simultaneously.

However, GPTs historically deliver what economists call a “level effect.” They make the economy permanently more productive, lifting GDP onto a higher path. They may also provide a modest “growth effect” by making innovation somewhat easier. But the long-run growth rate eventually returns to trend anchored by fundamentals, [including](#) the difficulty of discovery and the supply of human researchers.

History validates this framework for the GPTs of the past. Consider the [Industrial Revolution](#), [electrification](#), [computerization](#)—each produced massive one-time jumps in living standards, then growth rates settled.

The Explosive Case: AI Will Produce Ideas

Economists [Philippe Aghion](#) and [Simon Bunel](#), along with the [Epoch AI team](#), ask a different question: what if AI automates not just the production of goods but the production of *ideas*?

Standard growth theory treats ideas as a special input to production. A drug design or chip architecture is non-rivalrous—once it’s discovered, everyone can use it simultaneously, without depletion. Long-run growth comes from accumulating these non-rivalrous ideas. More ideas, more growth.

AI might alleviate the human bottleneck on idea production. But idea production has always been constrained by the supply of human researchers. More scientists mean more discoveries. Better scientific tools—microscopes, computers—help, but ultimately humans do the research, and human researchers scale with population.

AI breaks this association. If AI systems can conduct literature reviews, generate hypotheses, design experiments, and interpret results, then idea production decouples from demography. The research workforce scales with compute, not population.

The mathematical consequences of AI generating ideas could be explosive. If the number of researchers grows rapidly because AI researchers can be produced by other AIs—and if that scale overwhelms the rate at which good ideas become harder to find— GDP growth becomes super-exponential. Year one: 2% growth. Year five: 8% growth. Year ten: 25% growth. Economists call this an “economic singularity.”

Acceleration in idea-production holds clues to AI’s true potential. Is AI accelerating the rate of discovery? Evidence is intriguing but far from conclusive.

ML to navigate vast design spaces that were previously intractable.

Yet patents per researcher in the United States—a rough proxy for idea productivity—show no obvious inflection. Scientific publications are becoming steadily less disruptive. Drug approval timelines have not shortened. Time from hypothesis to breakthrough in major fields remains stubbornly long.

The seeming lag could be attributable to measurement error, since scientific impact takes years to manifest in data. Or the lag could be real—AI is helping at the margins but not transforming the core issue, which is that hard problems are hard.

Empirical trends in science will reveal whether AI is accelerating ideation. Understanding AI's impact on ideation is straightforward: watch science itself. If methods sections begin systematically crediting AI for hypothesis generation or experimental design, if patents per R&D dollar spent start rising, if time from research initiation to first patent filing compresses, if clinical trial success rates inflect upward, if venture-funded biotech suddenly posts dramatically higher hit rates, those findings suggest an active feedback loop. The key threshold is not whether AI assists researchers but whether research output begins scaling with compute investment rather than head count.

If scientific productivity continues its long-run decline, the standard pattern of GPT frameworks is likely to hold: we will get richer once, but we will not escape the gravity of diminishing returns.

Indicators to Resolve These Questions

longer arguments. They will be resolved by reality.

Right now, the skeptics of an outsized impact of AI on GDP are winning on points. Meaningful adoption is slower than headlines suggest. Hard tasks remain hard. Institutional friction in industries like health care is significant. Data from the US Census Bureau shows AI implementation below 10% in most sectors. Occupational wage gaps are not compressing. Scientific productivity shows no upward inflection.

Skeptics may be right—but only for now. Those predicting explosive growth are not making a claim about 2026. They are making claims about trajectories—some betting that compute scaling will allow for greater task complexity, others arguing that capital markets will overwhelm institutional friction, still others envisioning that recursive innovation will compound the pace of scientific discovery. These trajectories diverge in mechanism and timeline, but all predict that the constraints currently limiting AI's incursion into idea production will fall away. This absence of evidence is not evidence of absence, particularly for phenomena with long lag times.

Policymakers must work with the data they have. Leaders cannot wait for the debate to settle before acting, nor can they pretend to know which forecast is correct. But they can stop treating economic projections as prophecies to be believed or dismissed, and start treating them as hypotheses generating testable predictions.

The metrics proposed here are not the only relevant ones, but they may be sufficient to distinguish between the competing models. The Bureau of Labor Statistics tracks occupational wages. The Patent Office records filing timelines. Census surveys measure adoption depth. For these indicators, **we do not have a data problem; we have an attention-allocation problem.** The data is collected but not systematically analyzed as diagnostic indicators of which economic model is holding.

response is not to average the extremes, pick a camp, or wait for certainty. It is to build monitoring infrastructure that tells us which mechanisms are actually operating in the economy, then adapt in real time as evidence accumulates.

The quadrillion-dollar disagreement will resolve in time. The question is whether policymakers will be watching when it does—and whether we will have built the state capacity to respond to whichever world we are entering.

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